Question

Two identical samples of a common compound **A** (relative molecular mass not exceeding 80) found in daily life are boiled until completely evaporated, leaving no solid residue. The gas obtained from the first sample is passed sequentially through anhydrous calcium chloride and soda lime, with the latter increasing in weight by $33.0\ \mathrm{mg}$; the gas obtained from the second sample is passed sequentially through soda lime and anhydrous calcium chloride, with the latter increasing in weight by $25.5\ \mathrm{mg}$. Under an Ar atmosphere, $5.0\ \mathrm{mL}$ of a $2\ \mathrm{M}$ $\mathrm{MCl_3}$ solution is mixed with a large excess of an ethanol solution of **A**, immediately producing a precipitate. After filtration and drying, a purple fluffy powder **B** $5.15\ \mathrm{g}$ is obtained. **B** is readily soluble in water, stable in air, and has a non-zero magnetic moment measured at room temperature. Please select the correct option from the following choices.

A. The relative molecular mass of A is 79

B. +4 is also a common valence of M

C. Aqueous solutions of $\mathrm{MCl_3}$ are typically colorless

D. The metal cations in **B** may be in a tetrahedral coordination environment

Answer

Correct Answer: B

Detailed Explanation

Anhydrous calcium chloride can absorb gases including H_2O and NH_3, while soda lime can absorb acidic gases such as H_2O and CO_2. From this, it can be inferred that during the experiment with the second sample, anhydrous calcium chloride only absorbed NH_3. By testing various acidic gases, it can be observed that during the experiment with the first sample, if the gas absorbed by soda lime is CO_2, the molar ratio of the two gases is an integer ratio:

$$(33.0 \div 44.01)/(25.5 \div 17.034) = 0.501 \approx 1/2$$

CHECKPOINT 1 PTS

$n(CO_2) : n(NH_3) = 1 : 2$

This implies that **A** can decompose into CO_2 and NH_3 in a $1:2$ ratio, so **A** could be ammonium carbonate $(NH_4)_2CO_3$ ($96\ g/mol$), ammonium carbamate $H_2NCOONH_4$ ($78\ g/mol$), or urea $CO(NH_2)_2$ ($60\ g/mol$). Considering that **A** is a common compound in daily life with a relative molecular mass not exceeding 80, ammonium carbamate and ammonium carbonate can be ruled out. Therefore, **A** is urea, with the chemical formula $CO(NH_2)_2$.

CHECKPOINT 1 PTS

The chemical formula of **A** is $CO(NH_2)_2$

The molar mass of **B** is:

$$M_B = \frac{5.15}{2 \times 5.0 \times 10^{-3}} = 515 \text{ g/mol}$$

CHECKPOINT 1 PTS

$M_B = 515$ g/mol

In an Ar atmosphere and ethanol solution, urea most likely only undergoes a coordination reaction with MCl_3. Assuming the molecular formula of **B** can be expressed as $MCl_3 \cdot nCO(NH_2)_2 \cdot mC_2H_5OH$, substituting the relative molecular masses of urea and ethanol and testing multiple values reveals that only when $n = 6$ and $m = 0$ does **B** have a plausible chemical formula: $Ti[CO(NH_2)_2]_6Cl_3$. This is also consistent with details such as $Ti(III)$ being easily oxidized (hence the need for an Ar atmosphere), the product appearing purple, and having a non-zero magnetic moment.

CHECKPOINT 1 PTS

The chemical formula of **B** is $Ti[CO(NH_2)_2]_6Cl_3$

From this, it can be concluded that option A "The relative molecular mass of **A** is 79" is incorrect, option B "$+4$ is also a common oxidation state of M" is correct, option C "Solutions of $\mathrm{MCl_3}$ are usually colorless" is incorrect, and option D "The metal cation in **B** may be in a tetrahedral coordination environment" is incorrect.

CHECKPOINT 1 PTS

Option B is correct